Dhruv Raajeev

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Education

Texas A&M University College Station, TX

Bachelor of Science in Computer Science *Expected May 2028*

- Coursework: Data Structures and Algorithms; Program Design and Concepts; Discrete Structures; Linear Algebra

Experience

Quantitative Developer Jan. 2026 – Present

Maroon Fund (Student-Managed Investment Fund, $70K+ AUM) *College Station, TX*

- Backtested a combined buy/sell signal pipeline (NVDA, AMZN, META) with day-by-day trade simulation and trade-log sanity checks, computing 20+ metrics (Sharpe, Sortino, max drawdown, win rate, profit factor)
- Extended the backtester with S&P 500 benchmark analytics (alpha, beta, volatility) behind a 3-source data fallback (Alpha Vantage, yfinance, CSV) and CLI flags, enabling reproducible runs
- Fixed a train/test data leak where the holdout overlapped labeled training dates, rebuilding validation with causal features and a clean 263-day holdout covered by 25 unit tests

Software Development Intern July 2025 – Nov. 2025

Arytic *Austin, TX (Remote)*

- Built a Python PDF resume parser (pdfminer, regex) that segments resumes into structured JSON, extracting contact fields and matching skills against a 220+ term taxonomy
- Engineered a semantic search pipeline in Pinecone with 384-dim BERT-based embeddings (all-MiniLM-L6-v2, Hugging Face), ranking resumes against job descriptions by cosine similarity
- Prototyped 3 matching pipelines (TF-IDF, keyword lexicon, embeddings) to compare candidate-to-role retrieval

Projects

Amber | *Python, FastAPI, React, TypeScript, Docker, Azure, MCP* Sep. 2026 – Present

- Built a full-stack AI infrastructure simulator that lets users sketch LLM system architectures and run simulated load, projecting p50/p99 latency, utilization, bottlenecks, and monthly cost before deployment
- Modeled LLM inference (continuous batching, KV-cache limits, speculative decoding) and agentic workloads with rate-limit retries; shipped an MCP server so AI agents (Claude Code, Cursor) can simulate a user's codebase
- Validated the discrete-event engine against M/M/1 and M/M/c queueing theory within 0.8% over 100 runs each; 400+ pytest and Vitest tests run in GitHub Actions CI to catch frontend/backend API mismatches

Open Source Contributions | *TypeScript, Python* July 2026 – Present

- Repainted the 5x8 bitmap K glyph in pxpipe (7.4k stars), an LLM token-cost reducer that renders context as images, cutting K-to-H misreads 98% (42 to 1) in the maintainer's A/B test at zero token cost
- Wrote pxpipe's merged chars-per-token calibrator; a least-squares fit over 77K production requests measured 2.3 chars/token vs. the assumed 4, recalibrating the cost gate from 4 to 3
- Capped the file-read tool in Alibaba's chaosblade LLM agent (6.5k stars) at a 256 KB byte-exact prefix with a truncation notice, preventing context overflow and runaway token cost

pkgMirage | *TypeScript, Cloudflare Workers, KV, D1, MCP* Oct. 2026 – Present

- Deployed a Cloudflare Workers edge service (KV, D1, Cron Triggers, Rate Limiting) that screens npm/PyPI packages before install via an MCP server and npm registry proxy, blocking 132 of 132 hallucinated names
- Built a static risk checker weighing install code, package age, and lookalike names; without the malware list it flagged 82 of 131 malicious packages and blocked 0 of 2,978 popular legitimate ones

Pneumonia Localization | *Python, TensorFlow, Keras, Google Colab* Dec. 2023 – July 2024

- Trained and compared CNN segmentation architectures (ResNet50V2, InceptionV3, MobileNet U-Net) with ImageNet transfer learning on RSNA chest X-rays, reaching ~97% accuracy and 0.76 validation IoU
- Preprocessed ~30,000 DICOM images with a custom data generator (flipping, interpolation, RGB channel stacking, resizing) to improve model robustness across architectures

Relevant Skills

Frameworks & Languages: Python, TypeScript, JavaScript, Java, C++, SQL, HTML/CSS, FastAPI, React, Node.js, PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, pandas, NumPy

Tools & Platforms: Cloudflare Workers (KV, D1), Git, GitHub, GitHub Actions, Docker, Azure, Vercel, Pinecone, Hugging Face, REST APIs, Model Context Protocol (MCP), pytest, Vitest, Jupyter, Google Colab, VS Code, Cursor, Claude Code